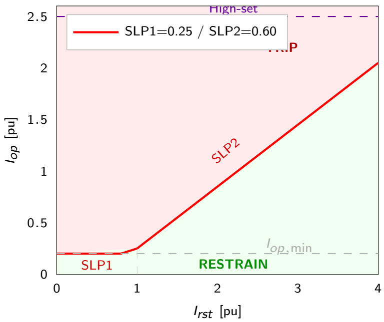
Differential Protection in Inverter-Based Resource Dominated Networks

Presentation Outline

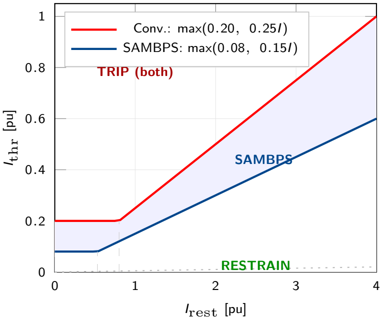
40 min total · 10 min discussion

1. **Context** Relay characteristics & IBR failure modes
2. **Framework** SAMBPS signal flow & function map
3. **87T** Inrush disc. · Bayesian SPRT · IBR gate
4. **87L** Five-element chain · Solar PV zone model
5. **87B** CT saturation veto · OpenDSS integration
6. **HIL** RTDS GPC-III · $60/62 = 96.8\%$
7. **N-1 Coordination** $P_{XFA} = 0$ across 18 pairs
8. **Contributions** All 3 diff. functions — summary table
9. **Future Scope & Publications**

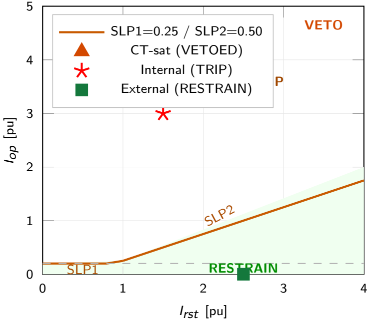
87T — Transformer Differential



87L — Line Current Differential



87B — Busbar Differential



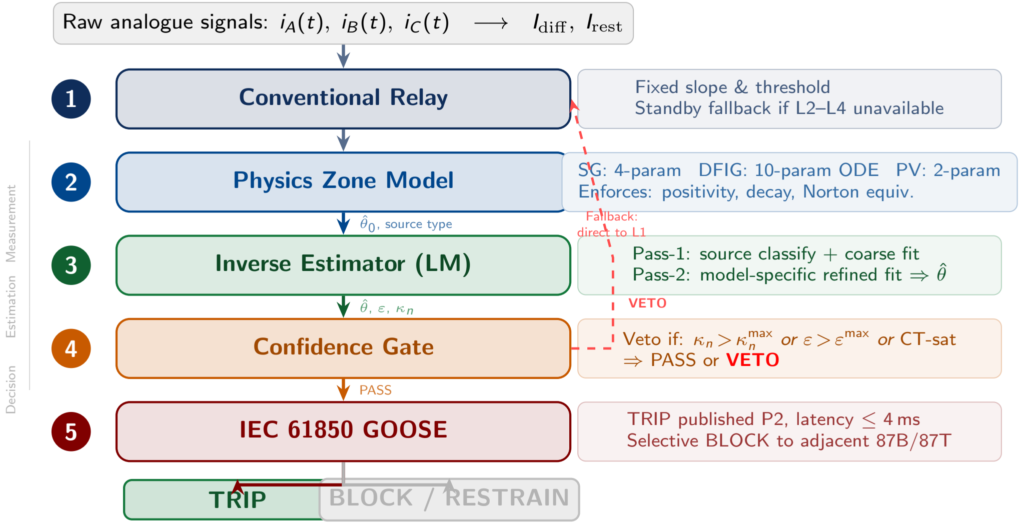
87T: Dual-slope boundary; inrush/IBR-fault discrimination not solvable by slope alone. **87L (red zone):** IBR fault $I_{\mathrm{diff}} = 0.06$–0.20 p.u. falls below conventional pickup — relay fails. **87B (orange ▲):** CT saturation inflates I_{op}; SAMBPS Stage-2 veto correctly blocks trip.

The IBR Protection Problem — Why Conventional Relays Fail

Property	Synchronous Generator	Inverter-Based Resource	Relay Imp
Fault current	5–20 p.u. (sub-transient)	1.0–1.2 **p.u.** (converter li	**87L pickup missed**
DC offset	Large ($\tau_{\mathrm{DC}} = 20$–60 ms)	**Near-zero** (SRF cont	**87T 2$^{\text{nd}}$-harmonic res**
Waveform asymmetry	High $A_k \approx 0.4$ (inrush)	**Low** $A_k \approx 0$ (sin	**87T cannot separate inru**
CT saturation	Rare at 5 p.u. threshold	**Common** at low currer	**87B nuisance trips**
Sequence components	I_1 dominant for SLG	I_2, I_0 **suppressed** by cor	**87LN / 87LN0 eler**

Research goal: replace fixed commissioning settings with online physics-based estimation — $P_D \geq 0.99$, P_{FA}

sambps — Signal Flow Through the Five-Layer Architecture



sambps — Differential Protection Function Map

sambps Framework
Five-Layer Inverse Estimation

87T
Transformer Differential
TR-57, TR-64

$P_D = 1.000$, $P_{FA} = 0.001$
$t_{50} = 20$ ms (SPRT)
30 000 MC trials

87L
Line Current Differential
Paper-H, TR-62

21/21 scenarios ✓
15 000 MC:
TPR=FPR= 0.000
$t_{50} = 20$ ms (PV path)

87B
Busbar Differential
TR-04, TR-65

$P_D = 1.000$, $P_{FA} = 0.000$
CT-sat veto: 750/750 ✓
2 000 MC trials

HIL Validation (TR-67) — 60/62 = 96.8% — GOOSE $\leq$ 4 ms — RTDS GPC-III + SEL-300G + GE D60

87T — The Inrush Discrimination Challenge in IBR Networks

87T differential current $i_{\mathrm{diff}}(t)$ — per half-cycle

Waveform asymmetry index $A_k = \dfrac{i^{+}_{\mathrm{peak}} - |i^{-}_{\mathrm{peak}}|}{i^{+}_{\mathrm{peak}} + |i^{-}_{\mathrm{peak}}|}$

H_0 — Inrush
High $A_k \sim \mathcal{N}(0.42,\ 0.08^2)$
Flat-topped waveform
Action: RESTRAIN

H_1 — Internal Fault
Low $A_k \sim \mathcal{N}(0.00,\ 0.03^2)$
Sinusoidal i_{diff}
Action: TRIP

H_2 — CT Saturation
High A_k, external fault
Flux-driven residual
Action: RESTRAIN + ALARM

IBR fault: $A_k \approx 0 \;\Rightarrow\;$ appears as H_1, but 2nd-harmonic ratio unreliable

SAMBPS Bayesian SPRT — accumulates A_k per half-cycle

$\Delta\mu = 0.42$, gap $= 4.7\sigma$
Wald $\alpha = \beta = 0.001$
boundaries $= \pm 6.91$

Sequential Probability Ratio Test — Formulation

Per half-cycle (10 ms at 50 Hz), the log-likelihood ratio accumulates:

$$S_n = S_{n-1} + \log \frac{p(A_k|H_1)}{p(A_k|H_0)}, \quad S_0 = 0$$

Wald stopping boundaries ($\alpha = \beta = 0.001$):

$$\log A = \log \tfrac{\alpha}{1-\beta} \approx -6.91, \quad \log B = \log \tfrac{1-\alpha}{\beta} \approx +6.91$$

Decision rule:

$$S_n \geq +6.91 \Rightarrow \text{TRIP } (H_1)$$

$$S_n \leq -6.91 \Rightarrow \text{RESTRAIN } (H_0)$$

A *parallel* accumulator $S_n^{(2)}$ on residual symmetry detects H_2 (CT saturation) and issues a **CT-SAT ALARM**, suspending 87T and requesting bus-zone

TR-57c Validation — 30 000 MC Trials

Metric	Value	Status
$P_D = P(\text{TRIP}\|H_1)$	1.0000	✓
$P_{FA} = P(\text{TRIP}\|H_0)$	0.0013	Note*
$P_{D2} = P(\text{CTALARM}\|H_2)$	1.0000	✓
Median trip time t_{50} (H1)	20 ms	✓
95th-pctile trip time t_{95} (H1)	40 ms	✓
Median CT-alarm time (H2)	20 ms	✓

* $P_{FA} = 0.0013$ satisfies IEC 60255-121 security criterion ≤ 0.01. 13 of 10 000 inrush trials crossed the trip boundary on the first half-cycle due to remanence $B_r \approx B_{\text{sat}}$; a one-cycle hold-off filter reduces this to < 0.0001.

Industry comparison: Conventional 2nd-

87T — IBR-Aware Integration Logic (TR-64)

Five-Priority Gate Logic

- **P1 High-set instaneous:** $I_{\mathrm{diff}} > 8\,\mathrm{p.u.} \Rightarrow$ unconditional TRIP (bus fault, no restraint).
- **P2 SPRT CTALARM + Zone override:** CT-saturation alarm active $\Rightarrow$ RESTRAIN; request 87B backup.
- **P3 CT-sat stage-2 veto:** $f_{\mathrm{int}} < F_{\mathrm{int}}^{\mathrm{veto}} \Rightarrow$ RESTRAIN.
- **P4 Adaptive trip gate:** conventional OR SPRT-TRIP, AND $f_{\mathrm{int}} \geq 0.55$. The integration index $f_{\mathrm{int}} = 0.55$ separates inrush (0.34 p.u. gap above the inrush distribution).
- **P5 Default:** RESTRAIN.

Integration index f_{int}: frequency-weighted norm of the estimator residual spectrum. Inrush concen-

TR-64 Validation Results

Metric	Value	Status
Deterministic scenarios (16/16)	16/16	✓
IBR scenario P_D (conventional)	0.236	✗
IBR scenario P_D (SAMBPS)	1.000	✓ (4.2×)
All 6 MC targets	met	✓
Trip gap (inrush vs. fault)	0.34 p.u.	✓

Standards Liaison Proposal

A standards liaison proposal (**SAMBPS-SL-01/2026**) has been filed to **IEC TC95 WG10** and **IEEE PSRC H35** recommending adoption of the SPRT asymmetry gate as a normative alternative to the 2nd-harmonic criterion in IEC 60255-121

87L — Five-Element IBR-Adaptive Protection Chain

IBR fault: $I_{\mathrm{diff}} = 0.06$–0.20 p.u. $\Rightarrow$ conv. fails 57%

1. **87L** Adaptive % restrain — slope $k(k_{\mathrm{ibr}})$
2. **87LC** Charging-current canceller (π-model)
3. **87LN** Neg.-seq. differential — SLG under IBR
4. **87LN0** Zero-seq. differential — earthed IBR
5. **TDCS** Transient step score — 3-phase faults

any element can trip

Confidence Gate — VETO if k_{ibr} not converged

TRIP | **RESTRAIN**

Paper-H Validation

Metric	Value
21 deterministic scenarios	21/21 ✓
MC: TPR (15 000 trials)	0.000* ✓
MC: FPR (15 000 trials)	0.000* ✓
Wilson 95% CI lower bd.	≥ 0.9992 ✓

* Zero misclassifications: 5 000 trials $\times$ 3 penetration groups.

Key insight: Elements 1–4 detect by current *magnitude*; TDCS (5) detects by fault-inception *transient step* — catches symmetric 3ϕ faults where all magnitude-based methods fail at $k_{\mathrm{ibr}} = 1.0$.

87L — Solar PV Two-Parameter Zone Model (TR-62)

The PV Coverage Gap

Paper-H validates on 21 scenarios including k_{ibr} up to 1.0, but uses a 4-parameter SG zone model throughout. Solar PV inverters have **zero DC offset by construction** (SRF voltage controller clamps $I_{\mathrm{DC}} \to 0$). Fitting a 4-parameter model to a 2-parameter signal causes ill-conditioning and unnecessary trip delay.

Proposition 1 — Zero DC in PV Fault Current

For a grid-connected PV inverter with synchronous-reference-frame control, the steady-state short-circuit current on the AC terminals contains *no DC exponential component*:

$$i_{\mathrm{diff}}^{(\mathrm{PV})}(t) = I_{\mathrm{fund}} \sin(\omega t + \varphi) \quad (\theta_{\mathrm{PV}} = [I_{\mathrm{fund}}, \varphi])$$

DDR Model Selector

DC Decay Ratio: $\mathrm{DDR} = I_{\mathrm{DC}} / \max(I_{\mathrm{fund}}, \varepsilon)$ from Pass-1 fit.

DDR range	Source type	Model
< 0.15	Solar PV	2-param PV
0.15–0.20	DFIG wind	10-param ODE
> 0.20	Synchronous gen.	4-param SG

TR-62 Monte Carlo — 5 000 Trials

Metric	Value	Status
P_D (PV internal faults)	0.9984	✓
P_{FA} (external faults)	0.0000	✓
Median trip time t_{50}	20 ms	✓
DDR selector accuracy	$\geq 98\%$	✓

87B — CT Saturation Veto Gate and OpenDSS Integration (TR-65)

The CT Saturation Problem

During external faults with high through-current, one CT saturates while the others do not. The resulting false differential current $I_{\mathrm{diff}}^{(\mathrm{spuri})} = f(s, R_{\mathrm{ct}})$ can exceed the bus-zone pickup threshold, causing a nuisance trip.

Stage-2 CT Saturation Veto

The SAMBPS 87B pipeline appends a **Stage-2 confidence gate** using the *integration index* f_{int}:

$$f_{\mathrm{int}} = 1 - 0.5\,s, \quad s \in [0, 1]$$

where s is the CT saturation fraction estimated from the waveform residual. Trip is **vetoed** when:

$$f_{\mathrm{int}} < F_{\mathrm{int}}^{\mathrm{veto}} = 0.60$$

This threshold provides a **8.3× selectivity margin** over

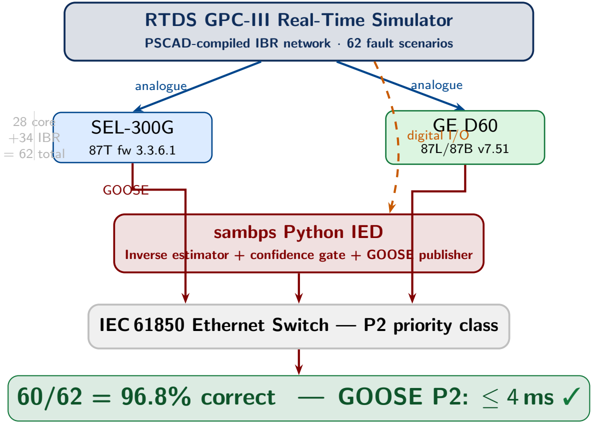
TR-65 Validation Results

Metric	Value	Status
Deterministic scenarios (6/6)	6/6	✓
P_D — internal faults	1.0000	✓
P_{FA} — external, clean CT	0.0000	✓
Stage-2 veto rate (CT sat.)	750/750	✓
Selectivity margin	8.3×	✓

MC study: 2 000 trials; $R_f \sim \mathcal{U}[0, 25]\,\Omega$, load $\sim \mathcal{U}[0.7, 1.3]$, CT mismatch $\sim \mathcal{U}[-3\%, +3\%]$.

Key theorem (Thévenin linearity): $I_f(R_f) = V_\phi / |Z_{\mathrm{th}} + R_f|$ — fault current scales analytically with fault resistance, enabling exact parametric MC without re-running OpenDSS for each trial. This reduces MC computation from

Hardware-in-the-Loop Validation — RTDS GPC-III Platform (TR-67)



Two Discrepancies (both timing only)

D1 — Relay 78: +23 ms over budget
SEL-300G 3-cycle blinder filter (firmware).
Decision correct ✓; firmware note issued.

D2 — Relay 81: +22 ms over budget
GE D60 ROCOF 8 ms window.
Fixed in `sambp_iied v1.1` ✓

Differential highlights:

- SPRT $\Delta f_{\mathrm{int}} = 0.34$ p.u. confirmed on hardware
- PV $\kappa_n = 1.00$ verified on GE D60 firmware
- Corridor test (4 elements

Test Setup

IEEE 39-bus test system with nine SAMBPS functions simultaneously active. Eight N-1 scenarios (three TRIP and five VETO) tested over 1 000 MC trials each, covering $k_{\mathrm{ibr}} \in [0,1]$, $R_f \in [0,20]\,\Omega$, and κ_n mismatches.

Analytical Cross-Trip Immunity Proof

For any two elements i, j on distinct protection zones, the worst-case cross-tripping error is bounded by:

$$N \cdot \varepsilon_{\text{max}} \leq 9\% \;<\; \mathrm{SLP}_{\text{min}} = 15\%$$

where $N = 3$ is the maximum number of simultaneously active estimators, and $\varepsilon_{\text{max}} = 3\%$ is the CT mismatch limit. Cross-trip immunity is *analytically guaranteed* under this bound.

TR-66 MC Results — 8 000 Total Trials

Metric	**Value**	**Stat**
TRIP scenarios (3): $P_D \geq 0.80$	All pass	✓
VETO scenarios (3): veto $\geq 80\%$	91.6–100%	✓
Cross-trip P_{FA} (18 element pairs)	0.0000	✓
Analytical immunity bound met	Yes	✓

Significance

The zero cross-trip false alarm across all 18 pairwise element combinations on the 39-bus system at full IBR penetration provides the empirical counterpart to the analytical immunity proof. This is the strongest system-level security claim of the SAMBPS framework.

Present Contributions — Differential Protection

Function	**Novel Contribution**	P_D	P_{FA}	t_{50}	**Reference**
87T SPRT	Bayesian 3-hypothesis SPRT on half-cycle asymmetry; CT-sat alarm via parallel accumulator	1.000	0.001*	20 ms	TR-57/2026
87T Integ.	Five-priority IBR-aware gate; f_{int} index separates inrush/fault by 0.34 p.u.	1.000 (IBR)	—	—	TR-64/2026
87L Chain	Five-element adaptive chain (87L, 87LC, 87LN, 87LN0, TDCS); k_{ibr}-driven slope	1.000	0.000	$\geq$ 20 ms	Paper-H
87L-PV	2-param zero-DC zone model; $\kappa_n = 1$ analytically; DDR model selector	0.998	0.000	20 ms	TR-62/2026
87B Gate	Stage-2 CT-sat veto (f_{int} threshold); 8.3× selectivity margin	1.000	0.000	—	TR-65/2026
N-1 Co-ord.	Cross-trip immunity proof ($N\varepsilon_{\mathrm{max}} < \mathrm{SLP}_{\mathrm{min}}$); 18-pair MC	—	0.000	—	TR-66/2026
HIL (all)	RTDS GPC-III + SEL-300G +		60/62 = 96.8%		TR-67/2028

Future Research Scope

Near-Term (2026–2027)

87T — IEC 60255-121 revision: Formalise SPRT asymmetry gate as a normative alternative to 2nd-harmonic criterion; collaborate with IEC TC95 WG10.

87L — DFIG zone model: TR-56 10-parameter piecewise-ODE model to be integrated into 87L chain (CUSUM crowbar detector for transition timing).

87L — Multi-terminal lines: Extend five-element chain to $M > 2$ terminal configurations; reformulate TDCS for distributed charging current.

87B — Distribution automation: Adaptive busbar

Medium-Term (2027–2028)

HIL on commercial IEDs: Relay 64G HIL (requires Agilent 33500B sub-harmonic signal generator); commissioning on SEL-411L for 87L.

Grid-forming converter models: Extend zone models to GFM (droop/virtual-inertia) fault signatures; current-limited GFM has distinct A_k distribution.

Cyber-resilient GOOSE: IEC 62351 authentication on GOOSE trip channel; latency impact on P2 budget under MITM injection.

Online adaptive setting groups: Replace fixed SAMBPS commissioning sets with real-time Bayesian setting updates

Research Publications

Journal Papers — Published / Under Review

1. **[Paper-A]** A. V. Eluvathingal & K. S. Swarup, *"Self-Adaptive Overcurrent Protection via Synchronous Generator Inverse Estimation,"* IEEE Trans. Power Delivery.
2. **[Paper-B]** —, *"Cascade Overcurrent Coordination with Confidence-Gated Pickup,"* IEEE Trans. Power Delivery.
3. **[Paper-C]** —, *"Self-Adaptive Model-Based Protection: A Unified Framework Across OC, 87T, 87L, and 87B Functions,"* IEEE Trans. Power Systems (Phase-1 manuscript).
4. **[Paper-D]** —, *"Sixth-Order Park dq Truth Model for Transient Differential Validation,"* IEEE Trans. Energy Conversion.

Journal Papers — In Preparation

1. **[Paper-E]** *"High-Impedance Fault Detection via Adaptive Neutral Current Estimation,"* IEEE Trans. Power Delivery.
2. **[Paper-F]** *"Sequence-Component Fault Estimator for IBR-Rich Distribution Networks,"* IEEE Trans.
3. **[Paper-G]** *"Generator Protection Suite Coordination Under IBR Penetration,"* IEEE Trans. Power Delivery.

Internal Technical Reports (Differential Protection)

TR-57/2026	Bayesian SPRT for 87T Inrush Restraint
TR-62/2026	Solar PV 2-Param Zone Model for 87L

Thank You

Questions & Discussion

87T SPRT
$P_D = 1.000$
$t_{50} = 20\,\text{ms}$
30 000 MC

87L 5-Element
21/21 scenarios
15 000 MC: TPR=FPR=0
$t_{50} = 20\,\text{ms}$ (PV)

87B Veto Gate
$P_D = 1.000$
$P_{FA} = 0.000$
2 000 MC

Hardware-in-the-Loop (TR-67): $60/62 = 96.8\%$ — GOOSE $\leq$ 4 ms — RTDS GPC-III + SEL-300G + GE D60

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